

Radiation Studies on the ATLAS Link and Control Daughterboard for the HL-LHC Upgrade

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Abstract—We present a summary of the radiation studies conducted on the latest version of the ATLAS Tile Calorimeter (TileCal) readout and control Daughterboard (DB), developed for the High Luminosity upgrade of the Large Hadron Collider. The DB functions as a interface hub between the on-detector and off-detector electronics by to handling configuration commands and synchronous timing signals, while simultaneously transmitting monitoring, calibration, and physics data from the front-end electronics. The design incorporates Commercial-off-the-shelf (COTS) devices with field-programmable gate arrays (FPGAs) and CERN custom radiation-hard GBTx ASICs. The COTS were thoughtfully studied and selected for their performance in radiation environments comparable to those expected over the operational lifetime of the collider. Each DB will provide continuous readout via two pairs of redundant 9.6 Gbps uplinks continuously receive LHC synchronous timing through a pair of redundant 4.6 Gbps downlinks. The upgraded calorimeter system will include 896 DBs serving a total of 5,376 high-speed optical links. A double-redundant architecture has been implemented to minimize single points of failure, while data integrity is preserved through techniques such as electronic redundancy layers, Triple Mode Redundancy (TMR), Soft Error Management (SEM) and the GBT protocol with Forward Error Correction (GBT-FEC). Extensive testing was performed to assess the impact of total ionizing dose (TID), non-ionizing energy losses (NIEL), and single-event effects (SEE) on system performance under realistic operating conditions.

Index Terms—Radiation Tolerant, TID, NIEL, SEU, SEL, SEE

I. INTRODUCTION

The ATLAS Tile Calorimeter (TileCal) is in charge of identifying hadronic jets and measuring transverse missing energy [1]. TileCal will be impacted by a harsher radiation environment and increased rates of pileup as a consequence of the High Luminosity Large Hadron Collider (HL-LHC) programme. An upgrade of the legacy calorimeter electronics is needed to guarantee full functionality over the 10 year period of the HL-LHC. The upgraded electronics will continuously transmit digitized data from approximately 10000 photo-

multiplier tubes (PMTs) to the off-detector systems. The on- and off detector electronics are interconnected via a radiation tolerant readout link and control Daughterboard (DB). Each DB is interfaced to a Upgrade TileCal Module that hosts up to 12 PMT channels in the front-end by means of a 400-pin FMC connector. A total of 896 DB will transmit approximately 35 Tbps of data via 5376 optical links to TileCal off-detector Pre-Processor modules. The on-detector systems, which include the DB, has been carefully engineered with redundancy layers and radiation mitigation strategies to maintain performance and reliability with negligible impact on data taking during LHC runs.

II. THE DAUGHTERBOARD DESIGN

The DB is a read-out link and control board, designed for the upgrade of TileCal [2]. A set of commercial of the shelf components and the CERN GBTx custom rad-hard ASIC [4] was selected for the design to achieve good performance for a decade of full functionality in the relatively low radiation environment simulated at the geographical position of the DB in the ATLAS detector. The design follows-up a redundancy layer of two functionally-equivalent halves used on-detector in TileCal modules [1].

Each DB powers four SFP+ modules that drive two 4.8 Gbps downlinks and four 9.6 Gbps uplinks. A pair of CERN GBTx hard ASICs receive LHC synchronized clock signals and configuration commands through the downlinks. Meanwhile, a pair of Kintex Ultrascale (KU) FPGAs distribute clocks and configuration received from the GBTx to the front-end and transmits front-end data and monitoring signals through the uplinks. The ASICs, flash memories and FPGAs of the whole board can be remotely configured through the downlinks.

The xADCs of the KU FPGAs monitor temperatures, voltage stability and current from both the DB and the front-end to assure that the on-detector electronics, inaccessible during runs, is well protected against any over-current or Single Event